

Horizon-RIC: Projection-Enforced Safety for AI-Controlled Radio Access Networks

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Executive Summary

Learned control has crossed into live actuation of licensed spectrum, and vendors have opened their rApp ecosystems to one another, so operators will soon actuate spectrum through planners they neither wrote nor can inspect, while keeping sole liability. Existing assurance methods — constrained reinforcement learning, Lagrangian penalties — lower the probability of a violation without bounding it, and bind the argument to one trained artefact. Horizon-RIC moves the constraint out of the objective and into the topology. A projection operator, the Decision Safety Shield, sits between the planner and the southbound interface: the planner emits a proposal, never an action, and only the projected action exists downstream, so an out-of-licence emission is unrepresentable rather than unlikely. A fail-closed guard chain refuses any emission whose corrections were not audited, and every decision and refusal is appended to a hash-chained, Ed25519-signed store. The property survives retraining, vendor substitution and a hostile planner. The layer is exercised against third-party open-source O-RAN software on real sockets — 12 of 12 policies ENFORCED by the production ric-plt/a1 mediator — driven by ray-traced propagation for 4096 receivers. Across 8000 decisions, 4658 requested actions were out of licence and none passed the emission boundary; 2442 were demanded by the propagation itself rather than by any attack. We ask for funding against the work packages below, and for the thing funding cannot buy: measured environments through the Alliance's endorsed labs, a member's planner to shield, and a working-group route for the profile.

Keywords — AI-RAN assurance, O-RAN, action projection, safe reinforcement learning, spectrum compliance, tamper-evident evidence, regulatory liability

All code, gates and result files are public: <https://github.com/danielfoojunwei/preceptualai-horizon-ric>. The claim ledger at docs/CLAIM_LEDGER.md states, for every number in this proposal, which CI job re-executes it and what is not yet gated.

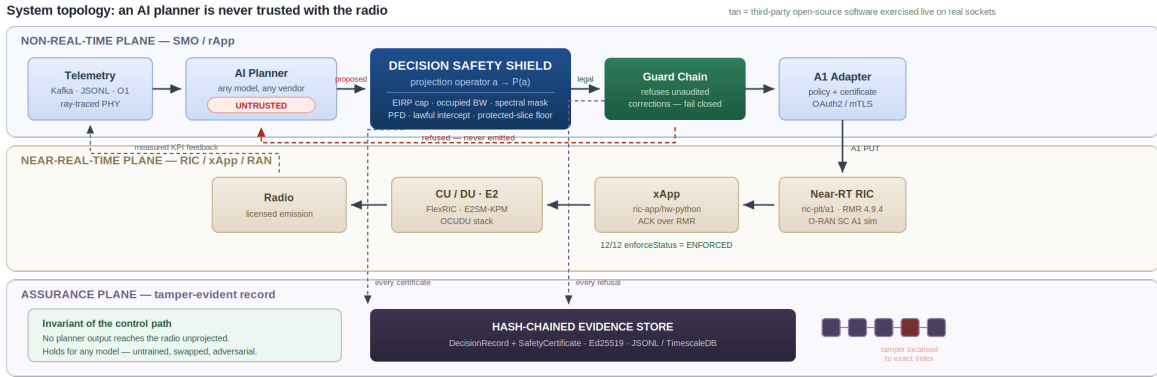


Fig. 1. Enforcement topology. The planner is untrusted: its output is a proposal, and only the projected action $P(a)$ leaves the non-real-time plane. The guard chain fails closed — an unaudited correction produces a refusal that never reaches the southbound interface. Southbound, the policy traverses third-party open-source software on real sockets. Every certificate and every refusal is appended to the assurance plane, where a single altered record is localised to its exact chain index.

I. Problem Statement and Market Relevance

A transmission that exceeds a licensed EIRP ceiling, or falls outside an assigned block, is a regulatory breach rather than a degraded KPI. Any assurance argument for AI in the RAN that terminates in a probability is therefore the wrong shape for the liability the operator carries. That mismatch, not model accuracy, is the practical reason AI is still deployed in advisory rather than actuating roles.

The industry has now crossed that line anyway. In Q1 2026 an AI uplink-interference rApp was field-trialled across roughly 1500 5G and 1300 4G live cells, lifting 5G SINR 27% under 10% higher uplink traffic, and was assessed at autonomy Level 3.86 on the path to Level 4 [9]. Level 4 removes the human from the per-decision loop. The same trial validated a third-party rApp over R1, and in March 2026 the two largest RAN vendors opened their SMO marketplace and rApp ecosystem to one another over the same interface [10]. The consequence is structural: operators will

increasingly actuate licensed spectrum through planners they did not write and cannot inspect, while retaining sole liability for the licence.

Market relevance. The buyer is the operator function that owns spectrum compliance and network assurance, and — symmetrically — the vendor that needs its rApp accepted into someone else's network. Two commercial facts fall out of our own measurements rather than from market assertion. First, the exposure is routine, not exceptional: on real campus geometry 46% of measured receivers require more than the licensed EIRP to close their link, so an unshielded planner breaches the licence on an ordinary day. Second, the control is not a performance tax: wherever the optimum is feasible, enforcement surrenders 0.0 dB, which is what makes it adoptable by the RAN team rather than merely mandated at them. We have no operator pilot and no revenue; converting technical readiness into commercial validation is precisely what WP3 and a member pilot are for, and we name it as the weakest part of this proposal rather than dress it up.

II. Innovative Solution and Technical Approach

Horizon-RIC takes the constraint out of the objective and puts it in the topology. A projection operator — the Decision Safety Shield — is interposed between the planner and the southbound interface. The planner's output is never an action; it is a proposal a , and only $P(a)$ exists downstream. An out-of-licence action is not penalised, it is unrepresentable at the emission point.

Three consequences follow that are architectural rather than empirical. First, the bound is worst-case rather than distributional and holds for an arbitrary planner, including a hostile one, because nothing in the argument refers to how the planner was trained. Second, it is auditable per decision: each emission carries a certificate naming the invariants that were violated, the corrections applied and the residual margins, appended to a hash-chained, Ed25519-signed store in which one altered record is localised to its exact index. Third, it is falsifiable: every published gate must be shown to fail under an injected regression, and a gate that cannot fail is treated as no gate.

The novelty is not shielding as an idea. Action projection for safe reinforcement learning is established prior art [4], [5], and we claim no priority over it. What is new is its realisation as a standards-facing enforcement plane for licensed spectrum: invariants written in the operator's regulatory vocabulary rather than in a reward function — EIRP ceiling, occupied bandwidth, spectral mask, power-flux density, lawful-intercept reachability, protected-slice capacity floor — composed to a fixed point, wrapped in a fail-closed guard chain, emitted over O-RAN A1/R1 as an ordinary rApp, and evidenced in a form a third party can replay without trusting us.

Enforcement path. Fig. 1 shows the decomposition. Telemetry enters the non-real-time plane; the planner proposes; the Shield projects; the guard chain refuses anything whose corrections were not audited; the A1 adapter emits the policy together with its certificate under OAuth2 or mTLS. Southbound, the policy crosses a near-real-time RIC into an xApp and onto the RAN. Certificates and refusals land independently in the assurance plane, so the record of what was refused does not depend on the component that refused it.

The Shield composes invariants, each supplying an evaluate and a project method, applied until a fixed point is reached so that satisfying one cannot silently violate another. They span regulatory limits and service commitments alike. A protected-slice floor, for example, reserves capacity that no planner can reallocate, restoring it pro rata so the allocation remains a valid simplex rather than a clipped vector; where the floor is unreachable the invariant deliberately stays unsatisfied so the guard chain refuses, rather than emitting an allocation that merely appears compliant.

Three distinct cryptographic mechanisms carry three distinct claims, and we keep them separate rather than calling them all signing: Ed25519 signs each per-decision safety certificate; RSA-PSS, optionally in an HSM over PKCS#11, signs model and dataset artefacts for provenance; RFC-3161 timestamps anchor the SHA-256 evidence chain to an external time authority.

Relation to the learning literature. This work sits on the dynamic spectrum access line developed by two of the present authors with Lam — federated deep reinforcement learning for IoT DSA [1], its hierarchical extension [2], and privacy-preserving spectrum pricing and power control for LEO satellite IoT [3]. That trajectory answers how well, and how safely, the learning itself can be conducted. What it does not address is the emission: an agent that selects a channel and a transmit power has produced an action some component must actuate, and securing or privatising the training pipeline does not constrain it. The relationship is complementary, not competitive — those agents decide which spectrum action is best; this layer ensures whichever is chosen is one the operator is licensed to emit.

III. Evidence Achieved to Date

Every result below is produced against software or data Horizon-RIC does not control, which is what makes it checkable, and each is re-executed by continuous integration on a clean runner — a committed result file is an assertion about a file, and only a gate makes it evidence about the system.

O-RAN interoperability. Policy emission was exercised against the O-RAN Software Community A1 simulator (12/12 accepted) and against the production Go A1 mediator `ric-plt/a1` with RMR 4.9.4 driving the official `ric-app/hw-python` xApp: 12/12 `enforceStatus = ENFORCED`, confirmed by a three-witness proof across emitter, mediator and xApp logs [13]. A workflow rebuilds RMR, the mediator and the xApp from source on a clean runner and re-asserts the result.

Enforcement now crosses both southbound interfaces, and is gated on both. The safety certificate rides on the A1 policy itself as an optional envelope carrying its digest, its Ed25519 signature and the invariants it violated; a CI job drives that across the vendored O-RAN Software Community A1 simulator on a real socket and verifies the signature against the returned digest. The control direction is gated the other way round: E2SM-RC control payloads are constructed from the O-RAN v1.03 standard ASN.1, and a decision the Shield refused cannot be encoded at all, so the near-real-time path cannot become a second unguarded route to the radio. Neither claim is stretched — the A1 result does not show a vendor RIC acting on the envelope, and the E2 result is construction and round-trip decode, not delivery.

Measured propagation. All benchmarks are driven by DeepMIMO [6] ray tracing of the ASU campus at 3.5 GHz — 4096 receivers, six 100 MHz subbands, per-path angular data — bound to a checksum-pinned archive that CI rebuilds and compares field by field. Provenance is enforced rather than asserted: perturbing the measurements must move a benchmark's outputs, and four benchmarks instead reject physically inconsistent input, having independently reconstructed gains from ray geometry.

Enforcement under real load. The decisive question is not whether zero out-of-licence actions were emitted but how many were attempted. Over 8000 decisions, each serving a measured receiver on its measured best subband at closed-loop power against its measured path gain, 4658 requests were out of licence and 0 passed the emission boundary (Fig. 2a). The composition matters more than the total: 2216 came from a poisoned planner, but 2442 came from the real geometry alone. A further 461 actions were inside the emission mask yet harmful by adjacent-channel leakage; after projection, 0 (Fig. 2b). Legality is adjudicated by an independent emission-mask and ACLR integral that does not share the Shield's constants, so the result cannot be an artefact of the system grading its own work.

Utility is not the price. Where the optimum is feasible, enforcement surrenders 0.0 dB. Because the projection maps every over-ceiling proposal onto the same executed action, the reward observed above the ceiling is constant, so a zero-data constant-power policy is already optimal there and measured learning gain is +0.000000; where an interior optimum exists the same learner gains +0.0096 to +0.0696. Both directions are enforced as gates, so no claim of AI benefit can be registered on a task incapable of showing one.

The same structure bounds an unrelated threat. White-box adversarial perturbation drives the neural receiver's symbol error up to 85 times the classical baseline, yet realised error stays inside a certified 1 dB envelope of that baseline for every attack in every regime, the worst case being 0.370 dB — without the layer having been coded against any of these attacks (Fig. 2d).

Integrity and robust aggregation. Traffic captured over the wire while ten real attacks were executed against a live free5GC

standalone core [8] is replayed through the artefact gate byte for byte: 11 of 11 probes detected or blocked, none bypassing. Against a model-replacement attack from 3 of 10 clients, FLTrust [7] attains 11.97 dB coverage RMSE against clean FedAvg's 11.92 dB — parity

— where Krum costs 14.46 dB. Under a Rényi accountant at a fixed certified $\epsilon = 4.1447$, calibrating the clip bound on public data moves utility from 340.12 dB to 14.34 dB without altering the guarantee: the same privacy certificate at 24 times less error.

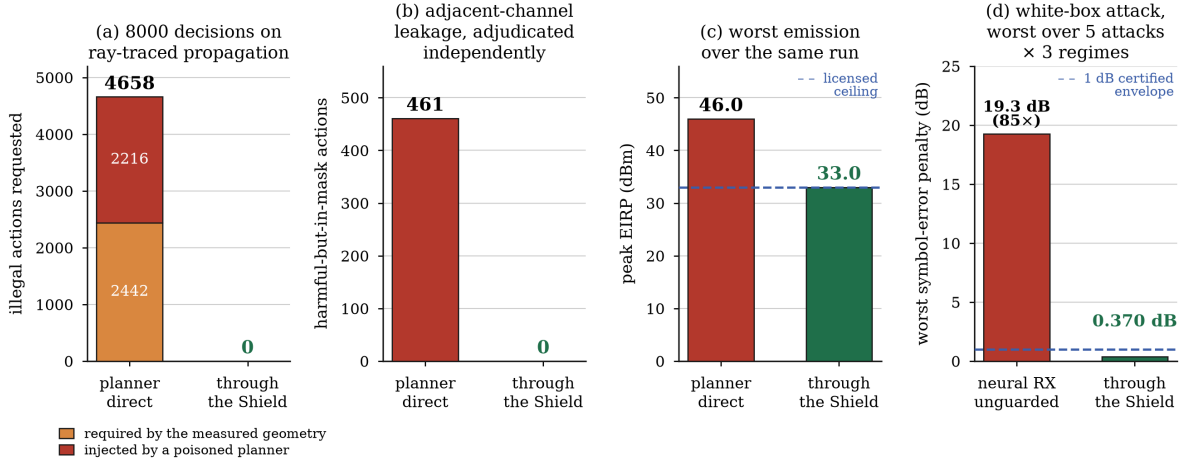


Fig. 2. Measured outcomes on ray-traced propagation, all four panels from the same CI-gated runs. (a) Of 4658 out-of-licence action requests over 8000 decisions, 2442 are demanded by the measured geometry alone and only 2216 by a poisoned planner — the exposure is an everyday one, not an attack artefact. (b) Actions inside the emission mask but harmful by adjacent-channel leakage, adjudicated by an oracle that does not share the Shield's constants. (c) Worst emission over the same run against a 33 dBm licence. (d) Worst-case symbol-error penalty of the neural receiver relative to the certified classical baseline, over five attacks in three propagation regimes; the Shield holds it inside a 1 dB envelope. Scales are per-panel: counts and dBm never share an axis.

IV. Deployment Feasibility

Enforcement is transparent to the standard interface. The 12/12 ENFORCED result above was obtained without patching the mediator or the xApp: the layer changes what may be emitted, not how it is emitted, so an operator adopts it without renegotiating its southbound contracts.

The system ships as an ordinary non-real-time rApp: container images; a Helm chart whose rendered templates are asserted by tests; EIAP and MantaRay onboarding descriptors written against the publicly documented surface; OAuth2 and mTLS credential paths; and a TimescaleDB evidence backend live-tested to SQL-level tamper detection. It is torch-free and installs on stock Python with no accelerator. Enforcement itself costs tens of microseconds per candidate action, so it sits well inside a non-real-time control period with room to spare.

It is also exercised against the open stack it must coexist with: the O-RAN SC A1 mediator and hw-python xApp [13], FlexRIC for E2SM-KPM and for the E2SM-RC ASN.1 this layer now encodes against, and the Linux Foundation's OCUDU CU/DU built from source with its E2 agent confirmed. Eight verification workflows gate every change; one rebuilds the licence-gated measurements from the pinned archive and re-verifies every published figure field by field.

What is not proven. No vendor platform has onboarded this — the EIAP and MantaRay packages are wire-contract tests against documented surfaces, not interoperability results. The RAN software is open-source rather than vendor equipment, the propagation is one scenario, and the OCUDU E2 join runs over a disclosed UDP transport substitution because the host kernel has no SCTP. The single external dependency this programme cannot supply for itself is access to a measured environment, which is why we ask the Alliance for it directly.

V. Twelve-Month Timeline and Milestones

Fig. 3 gives the schedule, owners and exit gates. Each gate is a falsifiable criterion; where a gate fails, the failure is published.

WP1, months 1–4 — conformance profile. The invariant and safety-certificate schema are mapped onto O-RAN A1 and E2 and released as a draft assurance profile with machine-readable schemas. Gate G1: a second, independently written planner is shielded without modification to either the planner or the Shield, and the gate pins the SHA-256 of both so editing either side to make the numbers agree fails it rather than passing it.

WP2, months 3–6 — second scenario and band. An independent ray-traced scenario with genuine frequency selectivity, at a different carrier. Gate G2: every published gate reproduces without retuning any constant, or the discrepancy is published.

WP3, months 6–9 — measured coexistence. The modelled incumbent is replaced by a multi-cell or bench-measured interference source. Gate G3: learning gain over the zero-data constant-power policy is positive on measured data — the honest test of whether AI contributes anything in this regime.

WP4, months 8–12 — audit-grade export and a delivered E2 control action. Evidence export in a form a spectrum authority can verify offline; and the E2SM-RC payloads that WP1 already constructs carried over a real E2 termination to a node with a UE attached. Gate G4: a RIC Control Request derived from a signed certificate is accepted by an E2 node, and one derived from a refused decision cannot be constructed at all.

WP5, months 10–12 — independent replication and standardisation. The profile is submitted to the AI-RAN Alliance working group and to O-RAN WG11 as candidate input, and a third party re-executes every gate from the published archive. Gate G5: an external party reproduces the headline enforcement result without contacting us.

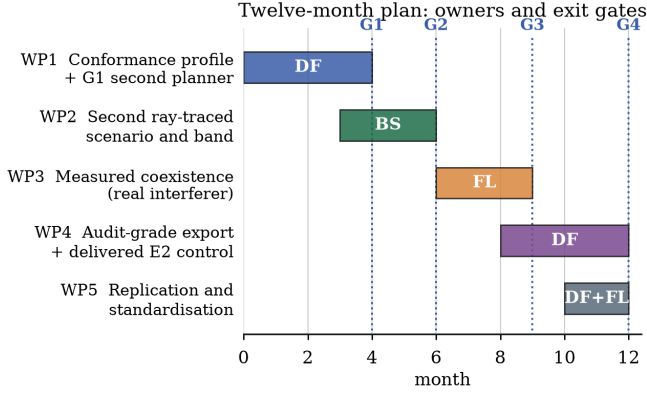


Fig. 3. Work packages, owners (DF Daniel Foo, BS Bowen Shen, FL Feng Li) and the five exit gates. Each gate is a falsifiable criterion, not a report — a gate that cannot fail is treated as no gate.

VI. Team, and What We Ask For

Daniel Foo (PreceptualAI, Singapore) is the architect and implementer of the enforcement layer, the evidence chain and the O-RAN integration, and leads WP1 and WP4. Bowen Shen (Nanyang Technological University) contributes the federated DSA and satellite-IoT spectrum work this layer sits on [1], [3] and leads WP2. Dr Feng Li (Nanyang Technological University) contributes the dynamic spectrum access and secure-learning programme [1], [2], [3] and leads WP3, with oversight of the measurement methodology across all packages. WP5 is shared between Daniel Foo and Dr Feng Li.

We ask for two kinds of support, and Table I separates them because they are not interchangeable. The first is funding, for the engineering and measurement effort in WP2 through WP5 — person-months on the second scenario, the measured-coexistence campaign, the audit-grade export and the delivered E2 control action, together with the RF bench instrumentation those require, the compute behind them, and an independently commissioned replication of every gate. No figure is quoted here on purpose: the right amount depends on which lab and which measured environment the Alliance can open, and a number named before that is a guess dressed as a budget. A costed breakdown against these exact lines follows on request.

The second kind is access, which no amount of funding substitutes for, and it is precisely what the Alliance has spent two years assembling: endorsed labs with Keysight Technologies, Northeastern University, Singapore University of Technology and Design and VIAVI Solutions, created for data creation and benchmarking; a Data-for-AI initiative defining a data-management and collection pipeline across real-time systems and simulators; a Test Methodology initiative developing standardised evaluation metrics [15]; and an AI-for-RAN working group that is the natural home for an assurance profile [14].

The asks are ordered by how much each changes the evidence rather than by how easy it is to grant. A measured interference source converts our weakest input — a modelled incumbent — into a measured one, and it is the only gate in the plan we cannot attempt alone. A member's planner converts G1 from an in-house demonstration into an industrial result: the layer never reads the planner, so a member risks no disclosure by letting us try, and a failure would be the most useful outcome in the programme because it would name an invariant the profile is missing. An operator willing to run the layer in shadow mode — certificates and refusals emitted, nothing actuated — would close the one criterion on which we are honestly weak.

Execution risk. The dominant risk is WP3: if no measured interference source or bench slot can be secured, gate G3 cannot be

attempted. The mitigation is sequencing — WP1, WP2, WP4 and WP5 carry no external dependency and deliver the profile, the second scenario, the export format and the replication regardless — plus a documented fallback to a multi-cell ray-traced incumbent, which we would publish as the weaker result it is rather than present as measured. The second risk is that a member's planner exposes an invariant our set cannot express; that outcome is a publishable finding about the profile's coverage, and the schema is versioned so it can absorb it.

Table I. What we ask for — funding against named work, and the access funding cannot buy — with what goes back in return. Amounts are in the costed breakdown, not here.

FUND Measured-coexistence campaign (WP3)	Person-months and instrumentation to replace the modelled incumbent with a multi-cell or bench-measured interference source — the weakest input in our evidence base and the only gate we cannot attempt alone. In return: the scripts, the pinned manifests and the raw results, published whether or not they favour us.
FUND Second ray-traced scenario and band (WP2)	Person-months on scenario generation and re-execution of every gate at a different carrier with genuine frequency selectivity. In return: a reproduction report naming any constant that turns out to have been tuned rather than derived.
FUND Audit-grade export and a delivered E2 control action (WP4)	Person-months to make the evidence chain verifiable offline by a spectrum authority, and to carry the E2SM-RC payloads WP1 already constructs over a real E2 termination. In return: the export format and the E2 binding, open and offered to the profile.
FUND Independent replication and security review (WP5)	An externally commissioned re-execution of every gate and a review of the fail-closed guard chain by someone with no stake in the answer. A result nobody outside the project has reproduced is not yet evidence, and we would rather fund that than assert it.
ACCESS Bench time in an endorsed lab	Funding alone does not buy this. The endorsed labs exist for data creation and benchmarking [15]; WP3 needs one of them, with a real interference source and calibrated instrumentation. In return: every measurement artefact, and co-authorship of what it produces.
ACCESS A member's planner to shield, under NDA or in the open	Turns G1 from a delivered demonstration into an industrial result. The layer never reads the planner, so a member discloses nothing by letting us try, and a failure is the most useful outcome available: it names an invariant the profile is missing. In return: the member gets a per-decision compliance record for its own planner, at no change to it.
ACCESS An operator willing to run the layer in shadow mode	Certificates and refusals emitted, nothing actuated. Closes the one criterion on which this proposal is honestly weak — no operator pilot. In return: an audit-grade evidence chain over the operator's own control path, and any defect we find reported before it matters.
ACCESS Data-for-AI alignment and a working-group route	Our results are bound to checksum-pinned archives and our gates fail when the input moves; that discipline is directly reusable as a provenance layer for Alliance datasets [15]. In return: the manifest and data-dependence tooling, the per-decision evidence chain as a candidate dataset format, the falsification rule offered to Test Methodology, and the profile submitted to AI-for-RAN [14].

VII. Expected Deliverables and Impact

Deliverables map onto the output types the Call anticipates. Prototype: the enforcement layer running as a non-real-time rApp with a live demonstration — policy emission through the production A1 mediator into a real xApp, including a Shield-corrected over-power proposal refused rather than emitted — which runs unattended and is available to the Alliance on request. Algorithms and models: the invariant algebra with its fixed-point composition, the protected-slice floor, and the certificate schema. Benchmarking-ready code: the harness in which every gate is proven to fail under injected regression, runnable by a third party from a pinned archive. Datasets: the pinned measurement manifests and the per-decision evidence chains, with the audit-grade export format from WP4. Standardization contributions: the invariant and certificate schema offered as a candidate assurance profile for enforceable service objectives, and the protected-slice floor as a concrete realisation of capacity an AI cannot reallocate.

Impact. The effect is to move a class of deployment blocker from statistical argument to architectural guarantee. An operator can adopt a stronger and less interpretable planner without enlarging regulatory exposure, because exposure is bounded by its own control path rather than by a supplier's model — and can demonstrate that

bound to a third party from the evidence chain rather than from validation statistics. A vendor gains a way to have its planner accepted into a network that will not inspect it. Published negative results are part of the deliverable, not an accident of it: the layer, schemas, evidence store and harness are developed in the open under a permissive licence with public CI, and every figure here is reproducible from a checksum-pinned archive.

VIII. Alignment with AI-RAN Alliance Priorities

The Alliance's problem is not whether AI can improve the RAN — the field trials have answered that — but what has to exist before an operator can let it actuate. This proposal supplies the missing artefact on the assurance side, and is deliberately shaped to be adoptable rather than merely publishable: vendor-neutral by construction, because the layer never reads the planner and so binds a member's model without that member disclosing anything about it; standards-facing, because the schema is offered as a profile rather than a product; and packaged for the exact integration path the March 2026 marketplace reciprocity created [10].

It also lands on the work the Alliance is already doing rather than beside it. The AI-for-RAN working group's remit is spectral and energy efficiency [14] — which is precisely where a learned planner wants to push against a licence, and therefore precisely where an enforcement layer earns its place. Test Methodology is developing standardised evaluation metrics for AI-driven RAN solutions [15]; our falsification rule, that every published gate must be shown to fail under an injected regression, is offered as a contribution to it, because a metric no one can fail is not a metric. Data-for-AI is building a structured collection pipeline across real-time systems and simulators [15]; the pinned-manifest and data-dependence tooling that binds our results to their inputs transfers directly, and the per-decision evidence chain is itself a candidate dataset format for assurance work.

The timing is set by the standards calendar. ITU-R M.2160 places security, privacy and resilience among the four design principles that apply to every IMT-2030 usage scenario and admits AI as a first-class scenario driver [11]; 3GPP opened 6G study in Release 20 with normative Release 21 work expected around end-2028 [12]. An assurance profile intended to be normative must exist, be implemented, and have survived falsification before that window closes. Table II maps this document onto the five official evaluation criteria.

Table II. The five official evaluation criteria, and where this proposal answers them.

1. Innovation and technical merit §II, §III — enforcement as a projection on the action space, with prior art [4], [5] conceded; every headline number re-executed by CI, and each gate proven to fail under injected regression.
2. Relevance to AI-RAN challenges §I, §VIII — addresses the blocker that keeps AI advisory rather than actuating, at the moment autonomy Level 4 [9] and R1 marketplace reciprocity [10] transfer planner authorship away from the operator.
3. Commercial impact and feasibility §I, §IV — 0.0 dB utility cost where the optimum is feasible; ships as an ordinary rApp with Helm, OAuth2/mTLS and onboarding descriptors. Weakest criterion: no operator pilot and no revenue yet.
4. Strength of execution plan §V, §VI — five work packages with named owners and falsifiable exit gates, four of them free of external dependency, plus a stated dominant risk with its mitigation and fallback.
5. Alignment with AI-RAN priorities §VI, §VII, §VIII — open, permissively licensed, pre-competitive; the schema offered as a candidate assurance profile to AI-for-RAN [14], the falsification rule offered to Test Methodology and the provenance tooling to Data-for-AI [15], ahead of the Release 21 window [11], [12].

IX. Evidence Boundary and What Is Not Claimed

We state the boundary precisely, because a proposal that overstates it cannot be checked. The propagation is site-specific ray tracing, not over-the-air capture; it is a single scenario; the RAN software is open-source rather than vendor equipment; the incumbent in the coexistence analysis is modelled. Live-network, vendor-interoperability, RF-conformance and carrier-scale properties are not claimed. “No action reached the radio” would overstate what was tested; the correct form is that no non-compliant action passed the Shield's software emission boundary in the tested O-RAN control path. The structural claim holds within the enumerated invariants, correctly configured limits, trusted telemetry and the guarded adapter path — it does not cover a missing invariant, a misconfigured licence limit, compromised telemetry, or a bypass around the adapter. No 6G or IMT-2030 compliance is claimed or claimable: no such specification yet exists. Closing these gaps is what WP2 and WP3 are for.

What the work most needs from members is measured environments: multi-cell measurements, a real incumbent, or bench time. Results, including failures, would be co-published.

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